ISRO's third rocket launchpad by 2029

Key features explained

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The Indian Space Research Organisation (ISRO) has officially begun work on its Third Launch Pad (TLP) at the Satish Dhawan Space Centre in Sriharikota. With a targeted completion date of March 2029, this new launch facility is more than an infrastructure expansion, it's a strategic platform that supports next-generation rockets and crewed spaceflight as India steps into a new era of space exploration.

A full-scale infrastructure upgrade

Unlike the existing two launchpads, the Third Launch Pad is being designed from scratch to support ISRO's future launch vehicles. At the heart of this is the Next Generation Launch Vehicle (NGLV), a modular, heavy-lift rocket family that will serve missions ranging from satellite deployment to human spaceflight. With the capacity to carry payloads of up to 30,000 kilograms to Low Earth Orbit, this rocket requires a more advanced and robust infrastructure than what the older pads can provide. The TLP will also support upgraded versions of the LVM3, particularly those equipped with semi-cryogenic stages.

It will include a flame deflection system to handle the intense exhaust and vibration from larger engines. New fueling systems are being integrated to manage cryogenic and semi-cryogenic propellants, which are essential for next-gen rockets.

A foundation for human and lunar missions

Perhaps the most important aspect of the TLP is that it's enabling India's human spaceflight ambitions. The pad is being designed to meet the safety standards for crewed missions. It will support abort systems, escape protocols, and integration processes specific to human-rated rockets. As ISRO moves toward establishing a space station by 2035 and sending astronauts to the Moon in the 2040s, the pad will be the gateway for these historic missions.

Indian MSMEs and aerospace firms will contribute to everything from structural components to advanced fueling. This not only helps lower costs but also strengthens India's domestic space supply chain. The ₹3,984.86 crore project is part of a broader trend of indigenisation in India's space sector, reinforcing the country's position as a rising space power. d